Homework 1 Solution

Problem 1: Properties of the ℓ_2 norm

Recall $\| \mathbf{x} \| = \sqrt{\mathbf{x}^T\mathbf{x}} = \sqrt{\sum_i x_i^2}$.

Part (a): Homogeneity, $\| c\mathbf{x} \| = |c| \, \| \mathbf{x} \|$

$$\| c\mathbf{x} \|^2 = \sum_i \left(cx_i\right)^2 = c^2 \sum_i x_i^2 = c^2 \| \mathbf{x} \|^2 .$$

Taking non-negative square roots gives $\| c\mathbf{x} \| = \sqrt{c^2} \, \| \mathbf{x} \| = |c| \, \| \mathbf{x} \|$. ■

Part (b): Triangle inequality, $\| \mathbf{x} + \mathbf{y} \| \leq \| \mathbf{x} \| + \| \mathbf{y} \|$

Cauchy-Schwarz: $|\mathbf{x}^T\mathbf{y}| \leq \| \mathbf{x} \| \, \| \mathbf{y} \|$. Proof: if $\mathbf{y} = \mathbf{0}$ it is trivial. Otherwise, for any real t, $0 \leq \| \mathbf{x} - t\mathbf{y} \|^2 = \| \mathbf{x} \|^2 - 2t\,\mathbf{x}^T\mathbf{y} + t^2 \| \mathbf{y} \|^2$. Choose $t = \mathbf{x}^T\mathbf{y} / \| \mathbf{y} \|^2$ to get $0 \leq \| \mathbf{x} \|^2 - \left(\mathbf{x}^T\mathbf{y}\right)^2 / \| \mathbf{y} \|^2$, which is Cauchy-Schwarz.

Triangle inequality: expand and apply Cauchy-Schwarz,

$$\| \mathbf{x} + \mathbf{y} \|^2 = \| \mathbf{x} \|^2 + 2\,\mathbf{x}^T\mathbf{y} + \| \mathbf{y} \|^2 \leq \| \mathbf{x} \|^2 + 2 \| \mathbf{x} \| \, \| \mathbf{y} \| + \| \mathbf{y} \|^2 = \left(\| \mathbf{x} \| + \| \mathbf{y} \| \right)^2.$$

Both sides are non-negative, so taking square roots gives the result. ■

Problem 2: A symmetric matrix and orthogonality

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \left[-\tfrac{1}{3}, 1, -1\right]^T, \quad \mathbf{v} = [1, 1, 1]^T.$$

Part (a): Compute $\mathbf{u} = A\mathbf{x}$

$$u_1 = 2\left(-\tfrac{1}{3}\right) + 1 = \tfrac{1}{3}, \quad u_2 = -\tfrac{1}{3} + 2 - 1 = \tfrac{2}{3}, \quad u_3 = 1 - 2 = -1,$$

so $\mathbf{u} = \left[\tfrac{1}{3}, \tfrac{2}{3}, -1\right]^T$.

Part (b): Orthogonality of $\mathbf{u}$ and $\mathbf{v}$

$$\mathbf{u}^T\mathbf{v} = \tfrac{1}{3} + \tfrac{2}{3} - 1 = 0,$$

so $\mathbf{u} \perp \mathbf{v}$.

Part (c): Is A symmetric positive definite?

$A = A^T$, so A is symmetric. For positive definiteness use leading principal minors (Sylvester's criterion):

$$\Delta_1 = 2 > 0, \; \Delta_2 = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3 > 0, \; \Delta_3 = \det A = 2(4-1) - 1(2-0) = 4 > 0.$$

Check with eigenvalues: $\det(A-\lambda I) = (2-\lambda)[(2-\lambda)^2 - 2] = 0$, so $\lambda = 2,\ 2 \pm \sqrt{2}$, i.e. $\approx 3.414,\ 2,\ 0.586$, all positive.

Alternatively, $\mathbf{z}^T A\mathbf{z} = z_1^2 + (z_1+z_2)^2 + (z_2+z_3)^2 + z_3^2 > 0$ for $\mathbf{z} \neq \mathbf{0}$.

Conclusion: yes, A is symmetric positive definite.

Problem 3: Three tosses of a fair coin

Part (a): $P(HHH)$

Tosses are independent, so $P(HHH) = (1/2)^3 = \frac{1}{8}$.

Part (b): Exactly one head

The $2^3 = 8$ outcomes are equally likely; exactly $\binom{3}{1} = 3$ of them (HTT, THT, TTH) have one head. So

$$P(\text{one head}) = \binom{3}{1}\left(\tfrac{1}{2}\right)^3 = \frac{3}{8}.$$

Problem 4: Box with two fair coins and one two-headed coin

Let F = fair coin chosen ($P(F) = 2/3$), T = two-headed coin chosen ($P(T) = 1/3$), H = heads.

Part (a): $P(H)$

Law of total probability:

$$P(H) = P(H|F)P(F) + P(H|T)P(T) = \tfrac{1}{2}\cdot\tfrac{2}{3} + 1\cdot\tfrac{1}{3} = \tfrac{2}{3}.$$

Part (b): $P(T \mid H)$

Bayes' rule:

$$P(T|H) = \frac{P(H|T)P(T)}{P(H)} = \frac{1\cdot\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}.$$